

ADV: Functions (Adv), F1 Working with Functions (Adv)
Absolute Value, Reflections and Inequalities (Y11)

Teacher: Ms Najem

Exam Equivalent Time: 51 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Functions, 2ADV F1 2019 HSC 2 MC

What values of x satisfy $4 - 3x \leq 12$?

- (A) $x \leq -\frac{16}{3}$
- (B) $x \geq -\frac{16}{3}$
- (C) $x \leq -\frac{8}{3}$
- (D) $x \geq -\frac{8}{3}$

2. Functions, 2ADV F1 2007 HSC 1b

Solve $2x - 5 > -3$ and graph the solution on a number line. (2 marks)

3. Functions, 2ADV F1 2009 HSC 1c

Solve $|x + 1| = 5$. (2 marks)

4. Functions, 2ADV F1 2010 HSC 1d

Solve $|2x + 3| = 9$. (2 marks)

5. Functions, 2ADV F1 2006 HSC 1e

Solve $3 - 5x \leq 2$. (2 marks)

6. Functions, 2ADV F1 2011 HSC 1e

Solve $2 - 3x \leq 8$. (2 marks)

7. Functions, 2ADV F1 2017 HSC 11g

Solve $|3x - 1| = 2$. (2 marks)

8. Functions, 2ADV F1 2018 HSC 11b

Solve $1 - 3x > 10$. (2 marks)

9. Functions, 2ADV F1 2008 HSC 1d

Solve $|4x - 3| = 7$. (2 marks)

10. Functions, 2ADV F1 SM-Bank 41

Find the values of x for which $|x + 1| = 5$. (2 marks)

11. Functions, 2ADV F1 SM-Bank 42

Find the values of x for which $|x - 3| = 1$. (2 marks)

12. Functions, 2ADV F1 SM-Bank 43

Solve $|3x - 1| = 2$ (2 marks)

13. Functions, 2ADV F1 SM-Bank 44

Solve $|x - 2| = 3$. (2 marks)

14. Functions, 2ADV F1 SM-Bank 37

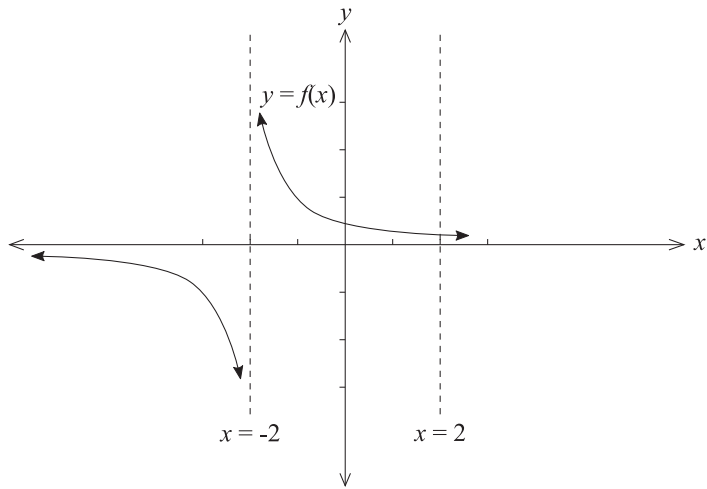
Find all values of x for which $|x - 4| = \frac{x}{2} + 7$. (3 marks)

15. Functions, 2ADV F1 SM-Bank 35

- i. Sketch the function $y = f(x)$ where $f(x) = (x - 1)^3$ on a number plane, labelling all intercepts. (1 mark)
- ii. On the same graph, sketch $y = -f(-x)$. Label all intercepts. (2 marks)

16. Functions, 2ADV F1 SM-Bank 36

Consider the function $f(x) = \frac{1}{x+2}$.



i. Sketch the graph $y = f(-x)$. (2 marks)

ii. On the same graph, sketch $y = -f(x)$. (2 marks)

17. Functions, 2ADV F1 2019 HSC 13e

(i) Sketch the graph of $y = |x - 1|$ for $-4 \leq x \leq 4$. (1 mark)

(ii) Using the sketch from part (i), or otherwise, solve $|x - 1| = 2x + 4$. (2 marks)

Worked Solutions

1. Functions, 2ADV F1 2019 HSC 2 MC

$$4 - 3x \leq 12$$

$$-3x \leq 8$$

$$x \geq -\frac{8}{3}$$

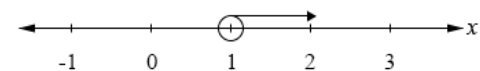
$$\Rightarrow D$$

2. Functions, 2ADV F1 2007 HSC 1b

$$2x - 5 > -3$$

$$2x > 2$$

$$x > 1$$



3. Functions, 2ADV F1 2009 HSC 1c

$$|x + 1| = 5$$

$$(x + 1) = 5 \quad - (x + 1) = 5$$

$$x = 4 \quad -x - 1 = 5$$

$$x = -6$$

$$\therefore x = 4 \text{ or } -6$$

4. Functions, 2ADV F1 2010 HSC 1d

$$|2x + 3| = 9$$

$$2x + 3 = 9 \quad - (2x + 3) = 9$$

$$2x = 6 \quad -2x - 3 = 9$$

$$x = 3 \quad -2x = 12$$

$$x = -6$$

$$\therefore x = 3 \text{ or } -6$$

5. Functions, 2ADV F1 2006 HSC 1e

$$3 - 5x \leq 2$$

$$-5x \leq -1$$

$$x \geq \frac{1}{5}$$

6. Functions, 2ADV F1 2011 HSC 1e

$$2 - 3x \leq 8$$

$$-3x \leq 6$$

$$x \geq -\frac{6}{3}$$

$$x \geq -2$$

7. Functions, 2ADV F1 2017 HSC 11g

$$|3x - 1| = 2$$

$$3x - 1 = 2 \quad \text{or} \quad -(3x - 1) = 2$$

$$x = 1 \quad \quad \quad -3x = 1$$

$$x = -\frac{1}{3}$$

$$\therefore x = 1 \text{ or } -\frac{1}{3}$$

8. Functions, 2ADV F1 2018 HSC 11b

$$1 - 3x > 10$$

$$-3x > 9$$

$$x < -3$$

9. Functions, 2ADV F1 2008 HSC 1d

$$|4x - 3| = 7$$

$$4x - 3 = 7 \quad \quad \quad -(4x - 3) = 7$$

$$4x = 10 \quad \quad \quad -4x + 3 = 7$$

$$x = \frac{5}{2} \quad \quad \quad -4x = 4$$

$$x = -1$$

$$\therefore x = \frac{5}{2} \text{ or } -1$$

10. Functions, 2ADV F1 SM-Bank 41

$$|x + 1| = 5$$

$$(x + 1)^2 = 5^2$$

$$x^2 + 2x + 1 = 25$$

$$x^2 + 2x - 24 = 0$$

$$(x + 6)(x - 4) = 0$$

$$\therefore x = 4 \text{ or } -6$$

11. Functions, 2ADV F1 SM-Bank 42

$$|x - 3| = 1$$

Method 1

$$(x - 3)^2 = 1$$

$$x^2 - 6x + 9 = 1$$

$$x^2 - 6x + 8 = 0$$

$$(x - 4)(x - 2) = 0$$

$$\therefore x = 2 \text{ or } 4$$

Method 2

$$(x - 3) = 1 \quad \quad \quad -(x - 3) = 1$$

$$x = 4 \quad \quad \quad -x + 3 = 1$$

$$x = -2$$

12. Functions, 2ADV F1 SM-Bank 43

$$|3x - 1| = 2$$

$$3x - 1 = 2 \quad - (3x - 1) = 2$$

$$3x = 3 \quad -3x + 1 = 2$$

$$x = 1 \quad 3x = -1$$

$$x = -\frac{1}{3}$$

$$\therefore x = -\frac{1}{3} \text{ or } 1$$

MARKER'S COMMENT: Note that both conditions must be satisfied! Dealing with negative signs and division for inequalities produced many errors.

13. Functions, 2ADV F1 SM-Bank 44

$$|x - 2| = 3$$

$$(x - 2)^2 = 3^2$$

$$(x^2 - 4x + 4) = 9$$

$$x^2 - 4x - 5 = 0$$

$$(x - 5)(x + 1) = 0$$

$$\therefore x = -1 \text{ or } 5$$

14. Functions, 2ADV F1 SM-Bank 37

$$x - 4 = \frac{x}{2} + 7 \quad \text{or} \quad -(x - 4) = \frac{x}{2} + 7$$

$$2x - 8 = x + 14 \quad -2x + 8 = x + 14$$

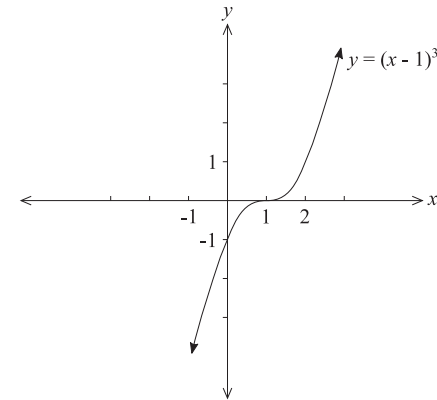
$$x = 22 \quad 3x = -6$$

$$x = -2$$

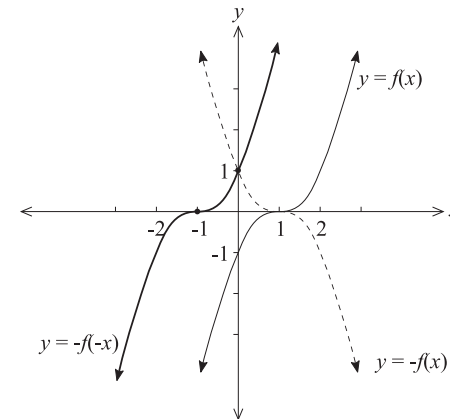
$$\therefore x = 22 \text{ or } -2$$

15. Functions, 2ADV F1 SM-Bank 35

i. $y = (x - 1)^3 \Rightarrow y = x^3$ shifted 1 unit to the right.



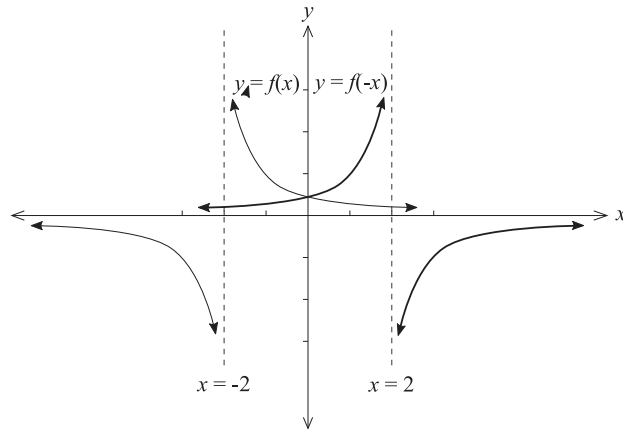
ii. $y = -f(x) \Rightarrow$ reflect $y = (x - 1)^3$ in x -axis.
 $y = -f(-x) \Rightarrow$ reflect $y = -f(x)$ in y -axis.



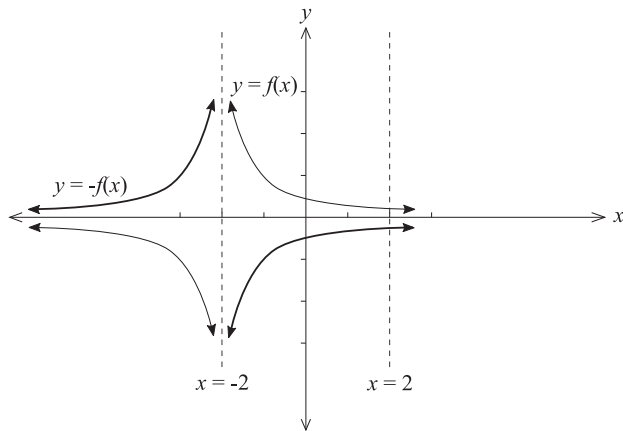
16. Functions, 2ADV F1 SM-Bank 36

i. Sketch $y = \frac{1}{x+2}$

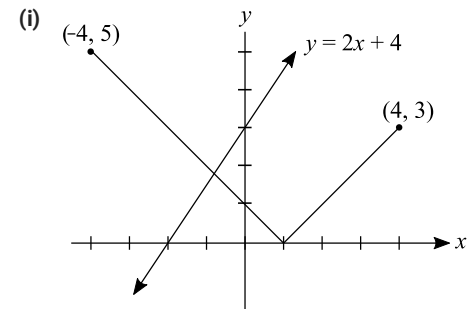
$y = f(-x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the y -axis.



ii. $y = -f(x) \Rightarrow$ reflect $y = \frac{1}{x+2}$ in the x -axis.



17. Functions, 2ADV F1 2019 HSC 13e



ii. By inspection, intersection when $x = -1$

Test:

$$|-1 - 1| = -2 + 4$$

$$2 = 2$$

$\therefore$ Intersection at $(-1, 2)$